

Kohsaku TOBIOKA

Publication List

- [1] Takaya Iwai, Ryosuke Sato, Kohsaku Tobioka, and Takumu Yamanaka. “Excluding MeV-scale QCD axions by $K_L \rightarrow \pi^0 \pi^0 a$ at KTeV” (Feb. 2026). To be published in JHEP. arXiv: [2602.15117 \[hep-ph\]](#).
- [2] Mitrajyoti Ghosh, Kevin Liguori, Takemichi Okui, and Kohsaku Tobioka. “Irreducible Bhabha background in the detection of muonium-antimuonium conversion”. *Phys. Rev. D* 113.5 (2026), p. 056014. DOI: [10.1103/5czk-1891](#). arXiv: [2510.25828 \[hep-ph\]](#).
- [3] Tae Hyun Jung, Takemichi Okui, Kohsaku Tobioka, and Jiabao Wang. “New bounds on heavy QCD axions from big bang nucleosynthesis”. *Phys. Rev. D* 113.5 (2026), p. 055002. DOI: [10.1103/12m1-h1cp](#). arXiv: [2510.23695 \[hep-ph\]](#).
- [4] Mitrajyoti Ghosh, Kevin Liguori, Takemichi Okui, and Kohsaku Tobioka. “Neutrino properties from muonium-antimuonium mixing”. *JHEP* 09 (2025), p. 144. DOI: [10.1007/JHEP09\(2025\)144](#). arXiv: [2504.05378 \[hep-ph\]](#).
- [5] Shameran Mahmud and Kohsaku Tobioka. “High energy vector boson scattering in four-body final states to probe Higgs cubic, quartic, and HEFT interactions”. *JHEP* 06 (2025), p. 153. DOI: [10.1007/JHEP06\(2025\)153](#). arXiv: [2501.16439 \[hep-ph\]](#).
- [6] Shameran Mahmud and Kohsaku Tobioka. “Energy growth in $V_L V_L \rightarrow V_L V_L, V_L V_L h$ scattering to probe Higgs cubic and HEFT interactions”. *JHEP* 09 (2024), p. 073. DOI: [10.1007/JHEP09\(2024\)073](#). arXiv: [2406.03522 \[hep-ph\]](#).
- [7] Kåre Fridell, Mitrajyoti Ghosh, Takemichi Okui, and Kohsaku Tobioka. “Decoding the $B \rightarrow K \nu \nu$ excess at Belle II: Kinematics, operators, and masses”. *Phys. Rev. D* 109.11 (2024), p. 115006. DOI: [10.1103/PhysRevD.109.115006](#). arXiv: [2312.12507 \[hep-ph\]](#).
- [8] Sudhakantha Girmohanta, Yuichiro Nakai, Yoshihiro Shigekami, and Kohsaku Tobioka. “Light dilaton in rare meson decays and extraction of its CP property”. *JHEP* 01 (2024), p. 153. DOI: [10.1007/JHEP01\(2024\)153](#). arXiv: [2310.16882 \[hep-ph\]](#).
- [9] Teppei Kitahara and Kohsaku Tobioka. “MeV sterile neutrino in light of the Cabibbo-angle anomaly”. *Phys. Rev. D* 108.11 (2023), p. 115034. DOI: [10.1103/PhysRevD.108.115034](#). arXiv: [2308.13003 \[hep-ph\]](#).
- [10] Yoav Afik, Babette Döbrich, Jan Jerhot, Yotam Soreq, and Kohsaku Tobioka. “Probing long-lived axions at the KOTO experiment”. *Phys. Rev. D* 108.5 (2023), p. 055007. DOI: [10.1103/PhysRevD.108.055007](#). arXiv: [2303.01521 \[hep-ph\]](#).
- [11] Teppei Kitahara and Kohsaku Tobioka. “Sterile neutrinos in light of the Cabibbo-angle anomaly”. *J. Phys. Conf. Ser.* 2446.1 (2023), p. 012009. DOI: [10.1088/1742-6596/2446/1/012009](#).
- [12] Kohsaku Tobioka. “Light new particles at the kaon experiments”. *J. Phys. Conf. Ser.* 2446.1 (2023), p. 012028. DOI: [10.1088/1742-6596/2446/1/012028](#).
- [13] Evgueni Goudzovski et al. “Weak Decays of Strange and Light Quarks” (Sept. 2022). arXiv: [2209.07156 \[hep-ex\]](#).
- [14] Mihoko M. Nojiri, Yasuihito Sakaki, Kohsaku Tobioka, and Daiki Ueda. “First evaluation of meson and τ lepton spectra and search for heavy neutral leptons at ILC beam dump”. *JHEP* 12 (2022), p. 145. DOI: [10.1007/JHEP12\(2022\)145](#). arXiv: [2206.13523 \[hep-ph\]](#).
- [15] Alexander Aryshev et al. “The International Linear Collider: Report to Snowmass 2021” (Mar. 2022). arXiv: [2203.07622 \[physics.acc-ph\]](#).

- [16] Evgueni Goudzovski et al. “New physics searches at kaon and hyperon factories”. *Rept. Prog. Phys.* 86.1 (2023), p. 016201. DOI: [10.1088/1361-6633/ac9cee](https://doi.org/10.1088/1361-6633/ac9cee). arXiv: [2201.07805](https://arxiv.org/abs/2201.07805) [hep-ph].
- [17] Emilie Bertholet, Sabyasachi Chakraborty, Vazha Loladze, Takemichi Okui, Abner Soffer, and Kohsaku Tobioka. “Heavy QCD axion at Belle II: Displaced and prompt signals”. *Phys. Rev. D* 105.7 (2022), p. L071701. DOI: [10.1103/PhysRevD.105.L071701](https://doi.org/10.1103/PhysRevD.105.L071701). arXiv: [2108.10331](https://arxiv.org/abs/2108.10331) [hep-ph].
- [18] Prateek Agrawal et al. “Feebly-interacting particles: FIPs 2020 workshop report”. *Eur. Phys. J. C* 81.11 (2021), p. 1015. DOI: [10.1140/epjc/s10052-021-09703-7](https://doi.org/10.1140/epjc/s10052-021-09703-7). arXiv: [2102.12143](https://arxiv.org/abs/2102.12143) [hep-ph].
- [19] Sabyasachi Chakraborty, Manfred Kraus, Vazha Loladze, Takemichi Okui, and Kohsaku Tobioka. “Heavy QCD axion in $b \rightarrow s$ transition: Enhanced limits and projections”. *Phys. Rev. D* 104.5 (2021), p. 055036. DOI: [10.1103/PhysRevD.104.055036](https://doi.org/10.1103/PhysRevD.104.055036). arXiv: [2102.04474](https://arxiv.org/abs/2102.04474) [hep-ph].
- [20] Adam Falkowski, Sanmay Ganguly, Phillippe Gras, Jose Miguel No, Kohsaku Tobioka, Natascia Vignaroli, and Tevong You. “Light quark Yukawas in triboson final states”. *JHEP* 04 (2021), p. 023. DOI: [10.1007/JHEP04\(2021\)023](https://doi.org/10.1007/JHEP04(2021)023). arXiv: [2011.09551](https://arxiv.org/abs/2011.09551) [hep-ph].
- [21] Sabyasachi Chakraborty, Tae Hyun Jung, Vazha Loladze, Takemichi Okui, and Kohsaku Tobioka. “Solar origin of the XENON1T excess without stellar cooling problems”. *Phys. Rev. D* 102.9 (2020), p. 095029. DOI: [10.1103/PhysRevD.102.095029](https://doi.org/10.1103/PhysRevD.102.095029). arXiv: [2008.10610](https://arxiv.org/abs/2008.10610) [hep-ph].
- [22] Stefania Gori, Gilad Perez, and Kohsaku Tobioka. “KOTO vs. NA62 Dark Scalar Searches”. *JHEP* 08 (2020), p. 110. DOI: [10.1007/JHEP08\(2020\)110](https://doi.org/10.1007/JHEP08(2020)110). arXiv: [2005.05170](https://arxiv.org/abs/2005.05170) [hep-ph].
- [23] G. Brooijmans et al. “Les Houches 2019 Physics at TeV Colliders: New Physics Working Group Report”. *11th Les Houches Workshop on Physics at TeV Colliders: Phys-Tev Les Houches*. Feb. 2020. arXiv: [2002.12220](https://arxiv.org/abs/2002.12220) [hep-ph].
- [24] Teppei Kitahara, Takemichi Okui, Gilad Perez, Yotam Soreq, and Kohsaku Tobioka. “New physics implications of recent search for $K_L \rightarrow \pi^0 \nu \bar{\nu}$ at KOTO”. *Phys. Rev. Lett.* 124.7 (2020), p. 071801. DOI: [10.1103/PhysRevLett.124.071801](https://doi.org/10.1103/PhysRevLett.124.071801). arXiv: [1909.11111](https://arxiv.org/abs/1909.11111) [hep-ph]. Selected as **PRL Editors’ Suggestion** and **Cover of the Issue**. Featured in *APS Synopsis*, *APS tip sheet*, *Science News*, *Phys.org*, *Popular Mechanics*, and *FSU News*.
- [25] Gordan Krnjaic, Gustavo Marques-Tavares, Diego Redigolo, and Kohsaku Tobioka. “Probing Muonphilic Force Carriers and Dark Matter at Kaon Factories”. *Phys. Rev. Lett.* 124.4 (2020), p. 041802. DOI: [10.1103/PhysRevLett.124.041802](https://doi.org/10.1103/PhysRevLett.124.041802). arXiv: [1902.07715](https://arxiv.org/abs/1902.07715) [hep-ph].
- [26] Xabier Cid Vidal et al. “Beyond the Standard Model Physics at the HL-LHC and HE-LHC” (2018). arXiv: [1812.07831](https://arxiv.org/abs/1812.07831) [hep-ph].
- [27] Xabier Cid Vidal, Alberto Mariotti, Diego Redigolo, Filippo Sala, and Kohsaku Tobioka. “New Axion Searches at Flavor Factories”. *JHEP* 01 (2019), p. 113. DOI: [10.1007/JHEP01\(2019\)113](https://doi.org/10.1007/JHEP01(2019)113). arXiv: [1810.09452](https://arxiv.org/abs/1810.09452) [hep-ph].
- [28] Alberto Mariotti, Diego Redigolo, Filippo Sala, and Kohsaku Tobioka. “New LHC bound on low-mass diphoton resonances” (2017). DOI: [10.1016/j.physletb.2018.06.039](https://doi.org/10.1016/j.physletb.2018.06.039). arXiv: [1710.01743](https://arxiv.org/abs/1710.01743) [hep-ph].
- [29] Walter Tangarife, Kohsaku Tobioka, Lorenzo Ubaldi, and Tomer Volansky. “Dynamics of Relaxed Inflation”. *JHEP* 02 (2018), p. 084. DOI: [10.1007/JHEP02\(2018\)084](https://doi.org/10.1007/JHEP02(2018)084). arXiv: [1706.03072](https://arxiv.org/abs/1706.03072) [hep-ph].
- [30] Walter Tangarife, Kohsaku Tobioka, Lorenzo Ubaldi, and Tomer Volansky. “Relaxed Inflation” (2017). arXiv: [1706.00438](https://arxiv.org/abs/1706.00438) [hep-ph].

- [31] Kfir Blum, Masazumi Honda, Ryosuke Sato, Masahiro Takimoto, and Kohsaku Tobioka. “ $O(N)$ Invariance of the Multi-Field Bounce”. *JHEP* 05 (2017). [Erratum: *JHEP*06,060(2017)], p. 109. DOI: [10.1007/JHEP05\(2017\)109](https://doi.org/10.1007/JHEP05(2017)109), [10.1007/JHEP06\(2017\)060](https://doi.org/10.1007/JHEP06(2017)060). arXiv: [1611.04570](https://arxiv.org/abs/1611.04570) [hep-th].
- [32] Itay M. Bloch, Rouven Essig, Kohsaku Tobioka, Tomer Volansky, and Tien-Tien Yu. “Searching for Dark Absorption with Direct Detection Experiments”. *JHEP* 06 (2017), p. 087. DOI: [10.1007/JHEP06\(2017\)087](https://doi.org/10.1007/JHEP06(2017)087). arXiv: [1608.02123](https://arxiv.org/abs/1608.02123) [hep-ph].
- [33] Ryosuke Sato and Kohsaku Tobioka. “LHC Future Prospects of the 750 GeV Resonance”. *Phys. Lett. B*760 (2016), pp. 590–593. DOI: [10.1016/j.physletb.2016.07.051](https://doi.org/10.1016/j.physletb.2016.07.051). arXiv: [1605.05366](https://arxiv.org/abs/1605.05366) [hep-ph].
- [34] Yuichiro Nakai, Ryosuke Sato, and Kohsaku Tobioka. “Footprints of New Strong Dynamics via Anomaly and the 750 GeV Diphoton”. *Phys. Rev. Lett.* 116.15 (2016), p. 151802. DOI: [10.1103/PhysRevLett.116.151802](https://doi.org/10.1103/PhysRevLett.116.151802). arXiv: [1512.04924](https://arxiv.org/abs/1512.04924) [hep-ph]. Featured in *APS Synopsis*, *Nature News*, *Physics World*, and *Phys.org*.
- [35] Kohsaku Tobioka, Ryuichiro Kitano, and Hitoshi Murayama. “Enhanced Higgs Mass in Compact Supersymmetry”. *JHEP* 04 (2016), p. 025. DOI: [10.1007/JHEP04\(2016\)025](https://doi.org/10.1007/JHEP04(2016)025). arXiv: [1511.04081](https://arxiv.org/abs/1511.04081) [hep-ph].
- [36] Gilad Perez, Yotam Soreq, Emmanuel Stamou, and Kohsaku Tobioka. “Prospects for measuring the Higgs boson coupling to light quarks”. *Phys. Rev. D*93.1 (2016), p. 013001. DOI: [10.1103/PhysRevD.93.013001](https://doi.org/10.1103/PhysRevD.93.013001). arXiv: [1505.06689](https://arxiv.org/abs/1505.06689) [hep-ph].
- [37] Kohsaku Tobioka. “A Natural Higgs Mass in Supersymmetry from Non-Decoupling Effects”. *PoS CORFU2014* (2015), p. 067.
- [38] Gilad Perez, Yotam Soreq, Emmanuel Stamou, and Kohsaku Tobioka. “Constraining the charm Yukawa and Higgs-quark coupling universality”. *Phys. Rev. D*92.3 (2015), p. 033016. DOI: [10.1103/PhysRevD.92.033016](https://doi.org/10.1103/PhysRevD.92.033016). arXiv: [1503.00290](https://arxiv.org/abs/1503.00290) [hep-ph].
- [39] Tomohiro Abe, Junji Hisano, Teppei Kitahara, and Kohsaku Tobioka. “Gauge invariant Barr-Zee type contributions to fermionic EDMs in the two-Higgs doublet models”. *JHEP* 01 (2014). [Erratum: *JHEP*04,161(2016)], p. 106. DOI: [10.1007/JHEP01\(2014\)106](https://doi.org/10.1007/JHEP01(2014)106), [10.1007/JHEP04\(2016\)161](https://doi.org/10.1007/JHEP04(2016)161). arXiv: [1311.4704](https://arxiv.org/abs/1311.4704) [hep-ph].
- [40] Xiaochuan Lu, Hitoshi Murayama, Joshua T. Ruderman, and Kohsaku Tobioka. “A Natural Higgs Mass in Supersymmetry from NonDecoupling Effects”. *Phys. Rev. Lett.* 112 (2014), p. 191803. DOI: [10.1103/PhysRevLett.112.191803](https://doi.org/10.1103/PhysRevLett.112.191803). arXiv: [1308.0792](https://arxiv.org/abs/1308.0792) [hep-ph]. Selected as **PRL Editors’ Suggestion**.
- [41] Ryosuke Sato, Satoshi Shirai, and Kohsaku Tobioka. “Flavor of Gluino Decay in High-Scale Supersymmetry”. *JHEP* 10 (2013), p. 157. DOI: [10.1007/JHEP10\(2013\)157](https://doi.org/10.1007/JHEP10(2013)157). arXiv: [1307.7144](https://arxiv.org/abs/1307.7144) [hep-ph].
- [42] Ryosuke Sato, Kohsaku Tobioka, and Norimi Yokozaki. “Enhanced Diphoton Signal of the Higgs Boson and the Muon $g-2$ in Gauge Mediation Models”. *Phys. Lett. B*716 (2012), pp. 441–445. DOI: [10.1016/j.physletb.2012.09.005](https://doi.org/10.1016/j.physletb.2012.09.005). arXiv: [1208.2630](https://arxiv.org/abs/1208.2630) [hep-ph].
- [43] Ryosuke Sato, Satoshi Shirai, and Kohsaku Tobioka. “Gluino Decay as a Probe of High Scale Supersymmetry Breaking”. *JHEP* 11 (2012), p. 041. DOI: [10.1007/JHEP11\(2012\)041](https://doi.org/10.1007/JHEP11(2012)041). arXiv: [1207.3608](https://arxiv.org/abs/1207.3608) [hep-ph].
- [44] Hitoshi Murayama, Yasunori Nomura, Satoshi Shirai, and Kohsaku Tobioka. “Compact Supersymmetry”. *Phys. Rev. D*86 (2012), p. 115014. DOI: [10.1103/PhysRevD.86.115014](https://doi.org/10.1103/PhysRevD.86.115014). arXiv: [1206.4993](https://arxiv.org/abs/1206.4993) [hep-ph].
- [45] Keisuke Harigaya, Shigeki Matsumoto, Mihoko M. Nojiri, and Kohsaku Tobioka. “Search for the Top Partner at the LHC using Multi- b -Jet Channels”. *Phys. Rev. D*86 (2012), p. 015005. DOI: [10.1103/PhysRevD.86.015005](https://doi.org/10.1103/PhysRevD.86.015005). arXiv: [1204.2317](https://arxiv.org/abs/1204.2317) [hep-ph].

- [46] Keisuke Harigaya, Shigeki Matsumoto, Mihoko M. Nojiri, and Kohsaku Tobioka. “Testing Little Higgs Mechanism at Future Colliders”. *JHEP* 01 (2012), p. 135. DOI: [10.1007/JHEP01\(2012\)135](https://doi.org/10.1007/JHEP01(2012)135). arXiv: [1109.4847](https://arxiv.org/abs/1109.4847) [[hep-ph](#)].
- [47] Hitoshi Murayama, Mihoko M. Nojiri, and Kohsaku Tobioka. “Improved discovery of a nearly degenerate model: MUED using MT2 at the LHC”. *Phys. Rev. D* 84 (2011), p. 094015. DOI: [10.1103/PhysRevD.84.094015](https://doi.org/10.1103/PhysRevD.84.094015). arXiv: [1107.3369](https://arxiv.org/abs/1107.3369) [[hep-ph](#)].

Also on [INSPIRE](#)